Pharmacy Sales Analysis & Inventory Recommendation Report

Prepared by Palak | Data: pharmacy_sales_v2.csv (1,000 transactions, 1 Jan – 27 Aug 2026, 5 stores, 6 medicines)

1. Dataset and Method

The sample dataset has 1,000 transactions across five stores (Bengaluru East, Chennai South, Delhi Central, Mumbai North, Pune West). It covers six medicines, one per drug class, each sold as Generic and Branded. There were no missing values, and every transaction's net revenue matches units × unit price × (1 − discount).

Profit was calculated as Net Revenue − (Units Sold × Unit Cost). Profit margin is Profit ÷ Net Revenue. All money is in INR.

Headline totals: net revenue ₹21,56,930; profit ₹8,27,117; overall margin 38.3%; 45,731 units sold.

Caveat: August covers only 1–27 August, so it is a partial month and its growth figures look weaker than they really are.

2. Monthly Revenue Growth by Drug Classification

Net revenue by month (₹ thousand):

Month	Analgesic	Antibiotic	Antidiabetic	Antihistamine	Antihypertensive	Lipid Lowering
Jan	16.1	74.5	31.7	24.5	25.3	115.2
Feb	20.6	103.4	17.0	20.0	37.1	79.6
Mar	12.0	83.6	18.7	31.8	45.3	43.3
Apr	14.0	91.7	22.5	29.1	39.1	82.9
May	16.3	55.8	28.6	11.6	72.6	59.3
Jun	21.8	75.9	39.7	23.6	46.2	93.6
Jul	15.4	96.3	35.7	27.8	31.9	77.9
Aug*	12.9	90.2	20.9	26.8	31.3	65.5

*Aug = 1–27 only.

Month-over-month revenue growth (%):

Month	Analgesic	Antibiotic	Antidiabetic	Antihistamine	Antihypertensive	Lipid Lowering
Feb	+28.2	+38.9	−46.5	−18.2	+46.9	−30.9
Mar	−41.7	−19.2	+10.1	+58.8	+22.1	−45.6
Apr	+16.8	+9.8	+20.6	−8.4	−13.8	+91.6
May	+16.4	−39.2	+27.3	−60.2	+85.7	−28.5
Jun	+33.3	+35.9	+38.5	+103.9	−36.4	+57.8
Jul	−29.5	+27.0	−9.9	+17.8	−30.9	−16.8
Aug*	−15.7	−6.3	−41.5	−3.5	−1.7	−16.0

What this shows:

- Comparing January with July (the last full month), Antibiotics grew 29.3% and Antihypertensives 26.2%. Antihistamines (+13.8%) and Antidiabetics (+12.8%) also grew. Analgesics were roughly flat to down (−4.5%), and Lipid Lowering fell sharply (−32.3%).
- Month-to-month swings are very large in every class, often ±40% or more. Any single month's growth figure is therefore a weak guide, and the trend over several months is more reliable.
- Antibiotics are the most stable high-revenue class, while Lipid Lowering is the most volatile large one.

3. Generic vs Branded Formulations

Formulation	Units sold	Net revenue (₹)	Profit (₹)	Margin
Generic	22,716	10,68,819	4,10,495	38.4%
Branded	23,015	10,88,110	4,16,621	38.3%

By medicine:

Medicine	Generic units	Branded units	Generic margin	Branded margin
Amlodipine 5mg	3,865	3,817	43.7%	44.1%
Amoxicillin 500mg	3,460	4,049	31.7%	31.8%
Atorvastatin 20mg	4,296	3,877	36.5%	36.4%
Cetirizine 10mg	3,565	4,502	50.7%	50.2%
Metformin 500mg	3,777	3,134	42.3%	41.8%
Paracetamol 650mg	3,753	3,636	42.7%	43.0%

What this shows:

- Generic and branded products perform almost identically, in both margin (38.4% vs 38.3%) and units sold. The unit-sales gap is not statistically significant (t-test p = 0.23).
- In this dataset, the generic and branded versions of each medicine have the **same unit price and unit cost**. The only difference is the discount applied, so the data does not show a real generic-vs-branded price advantage. Real-world generics are usually cheaper to buy and sell at a different margin, so this dataset likely does not reflect that. The finding should be read as "no difference in this sample", not as a general rule.
- Branded leads in units for Amoxicillin and Cetirizine, while Generic leads for Atorvastatin and Metformin.

4. Top 5 Revenue-Generating Medicines

Rank	Medicine	Net revenue (₹)	Units	Profit (₹)	Margin
1	Amoxicillin 500mg	6,71,425	7,509	2,13,376	31.8%
2	Atorvastatin 20mg	6,17,366	8,173	2,25,062	36.5%
3	Amlodipine 5mg	3,28,909	7,682	1,44,541	43.9%
4	Metformin 500mg	2,14,779	6,911	90,381	42.1%
5	Cetirizine 10mg	1,95,215	8,067	98,411	50.4%

Paracetamol 650mg is sixth (₹1,29,236 revenue, 42.8% margin). The dataset has only six medicines, so the top 5 leaves out just one.

Amoxicillin and Atorvastatin together make up about 60% of all revenue. Cetirizine has the best margin (50.4%) but ranks fifth on revenue because of its low unit price, and Amoxicillin has the lowest margin (31.8%).

5. Inventory Recommendations

These are suggestions based on 8 months of sales data. The dataset has no stock levels, lead times or expiry dates, so the actual quantities should be set with those inputs.

Demand reference (average monthly units, June–July): Metformin 1,216; Atorvastatin 1,141; Cetirizine 1,059; Paracetamol 1,053; Amoxicillin 958; Amlodipine 905.

Variability (coefficient of variation of monthly units, Jan–Jul): Amlodipine 0.36; Metformin 0.32; Atorvastatin 0.29; Cetirizine 0.27; Paracetamol 0.21; Amoxicillin 0.19.

1. **Protect Amoxicillin and Atorvastatin first.** They bring in about 60% of revenue, so stock-outs here cost the most. Amoxicillin demand is steady (lowest variability), so a lean, regular reorder cycle works. Atorvastatin demand is more variable, so it needs a larger safety stock.
2. **Hold extra safety stock for Amlodipine and Metformin.** Their monthly demand is the least predictable (Amlodipine peaked at 1,690 units in May against 588 in January). Smaller, more frequent orders with a buffer would reduce both stock-outs and overstock.
3. **Keep Cetirizine well stocked despite its lower revenue.** It has the highest margin (50.4%) and high unit volume, so each unit sold is worth more than for any other medicine.
4. **Order Paracetamol on a simple, steady schedule.** It is a low-value, moderately stable product that does not justify large safety stock.
5. **Match stock to store demand.** Mumbai North sells the most Cetirizine (2,311 units), Chennai South the most Metformin (1,906), and Bengaluru East the most Atorvastatin (1,987) and Paracetamol (1,866). Delhi Central leads on Amlodipine (1,849). Pune West has the lowest revenue (₹3,70,216) and should be stocked more conservatively.
6. **Do not decide on generic vs branded by margin alone.** The data shows no margin difference between them. Choose between them based on actual purchase cost, supplier terms and demand, which this dataset does not capture.
7. **Revisit when August is complete,** and re-check the Analgesic and Lipid Lowering trends, which are the two weakest classes.

6. Limitations

- Only 8 months of data, with August incomplete, so seasonal patterns cannot be confirmed.
- The sample appears synthetic: generic and branded prices and costs are identical, so some real-world effects are missing.
- There are no inventory, supplier or expiry data, so recommendations are directional.